

UNIQA Conversion Coach

Reads each customer, fires the **one** right nudge – or stays silent – and is **proven in simulation before it ever touches a real user**.

Live demo: qwadratic.github.io/uniqua-conversion-coach · self-contained engine, MIT

The funnel loses **94%** of buyers – and you can't A/B test on real ones.

5.6%

online conversion today

67%

gone at the first price screen

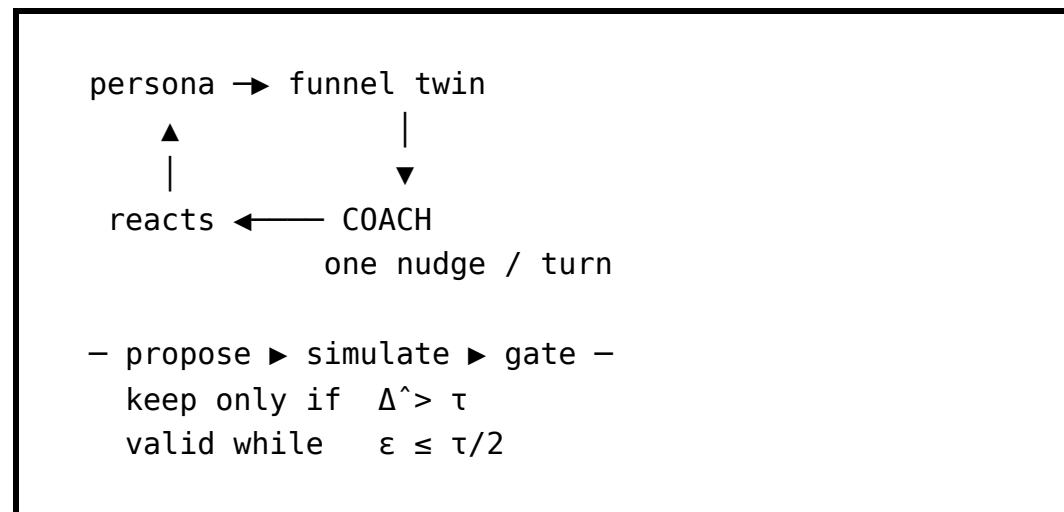
78%

gone at the final price

- › **One shot per visitor, regulated product** – any "fix" is a guess until it ships.
- › **One nudge can't fit all.** For one persona an advisor handoff is the win; for another it's a **failure**.

Don't rebuild the funnel. Simulate it, then coach it.

- › **3 LLM personas** click a faithful twin of the funnel – so we A/B in the sim, not on customers.
- › **The coach sees only a generic event log** (no health data) and fires one typed widget. Ports to **any** funnel.
- › **It improves itself**, gated by one number: $\epsilon = \text{distance}(\text{sim}, \text{reality})$. Keep ϵ small $\Rightarrow$ synthetic wins are real.



Conversion **doubled** in the sim. And the loop proves the win is real.

2×

conversion in the paired sim A/B
– coach on vs off, on the
calibrated persona mix.

–4.2pp

a bad coach the loop caught from
traces – and the gate rejected.
Wins are real, not noise.

1.5B

local LoRA adapters trained clean
after we fixed the distillation
collapse.

- › **Why the 2× is trustworthy:** personas are calibrated to the real funnel ($\epsilon=0.097$, anchors eval-only), and the gate ships a coach only when measured success rises – it already rejected one that didn't.

We proved the loop can't stop improving.

A theorem (on paper): while $\epsilon \leq \tau/2$, every coach the gate accepts is a **real**, monotone gain – so the workflow guarantees continuous self-improvement, not just synthetic wins.

Next: machine-check it in Z3 · wire production logs into Loop B · ship the distilled coach as one tagged local model.

One nudge. Right person. Right moment. **Proven before it ships.**

qadratic.github.io/uniqa-conversion-coach